# AIRCRAFT BOOKING SYSTEM

Web Application for Flight, Passenger, Seat and Booking Management

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# PROGRAMMING LANGUAGES

- Backend: JAVA
- Frontend: TypeScript
- Angular templates: HTML
- Styling: CSS

# ARCHITECTURE

Full-stack web application

Client-Server architecture

RESTful API communication

Component-based frontend architecture

Entity-Repository-Controller pattern

# BACKEND TECHNOLOGIES & TOOLS



Backend development:

**Spring Boot**



REST API endpoints/controllers:

**Spring Web**



Data access & management:

**Spring Data JPA**



ORM & entities:

**Hibernate/JPA**



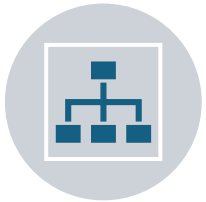
Passwords encryption:

**BCryptPasswordEncoder**



Validation annotations for entry control:

**Jakarta Validation**



Build & dependency management:

**Maven**

# FRONTEND TECHNOLOGIES & TOOLS



Frontend Framework:
**Angular**



Pages Navigation:
**Angular Router**



Backend API Communication:
**Angular HttpClient**



Forms:
**Reactive Forms**



UI Components:
**Angular Material**

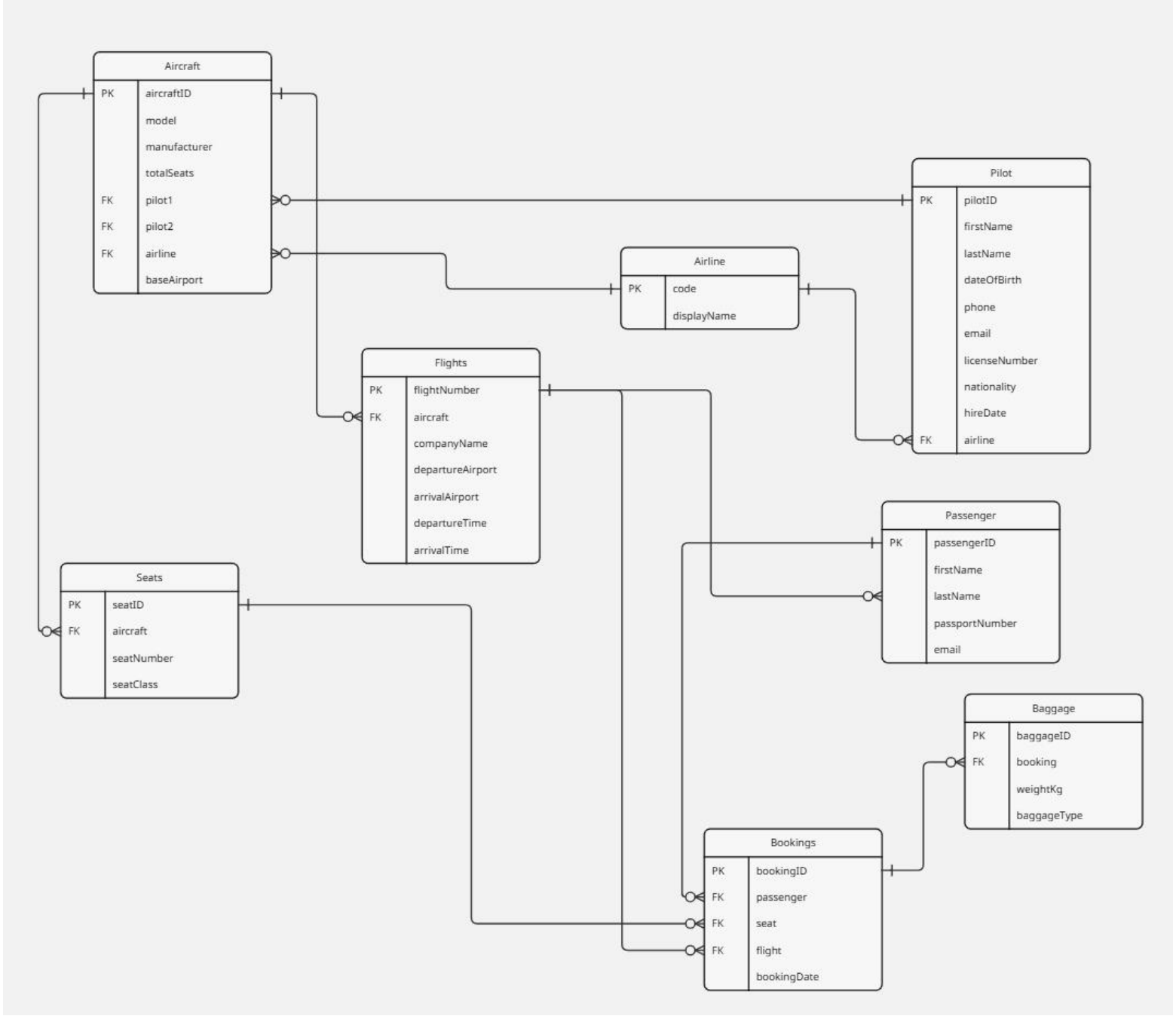


Angular Standalone Components

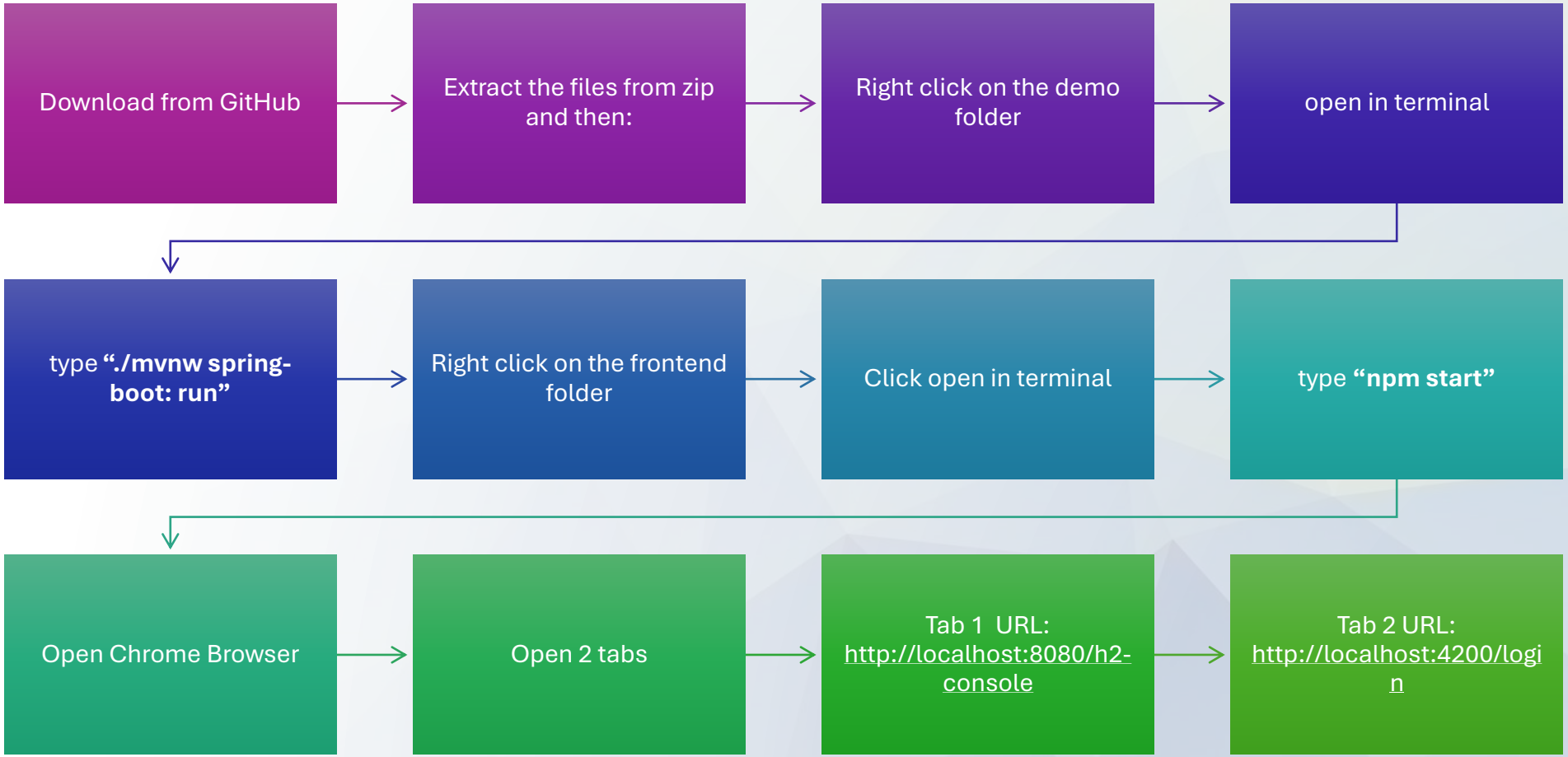
# APPLICATION SERVICES

- Authentication / Login
- Flight Management
- Passenger Management
- Booking Management
- Baggage Management
- Aircraft Management
- Seat Management
- Pilot Management
- Retrieve Available Seats per Flight
- Filter and Display Bookings / Flights
- Check Pilot and Seat Availability

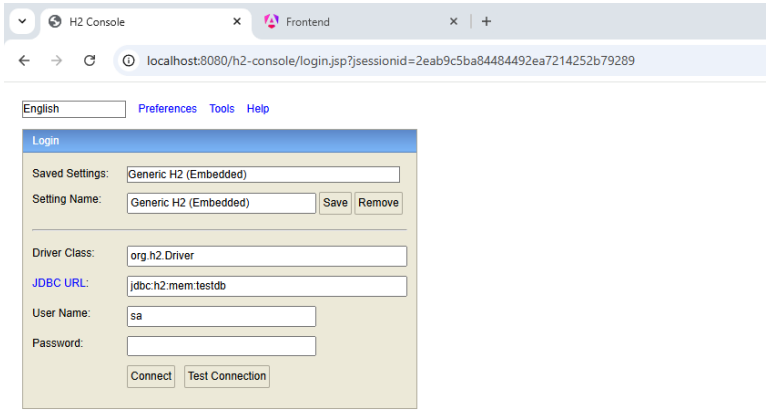
# CLASS DIAGRAM

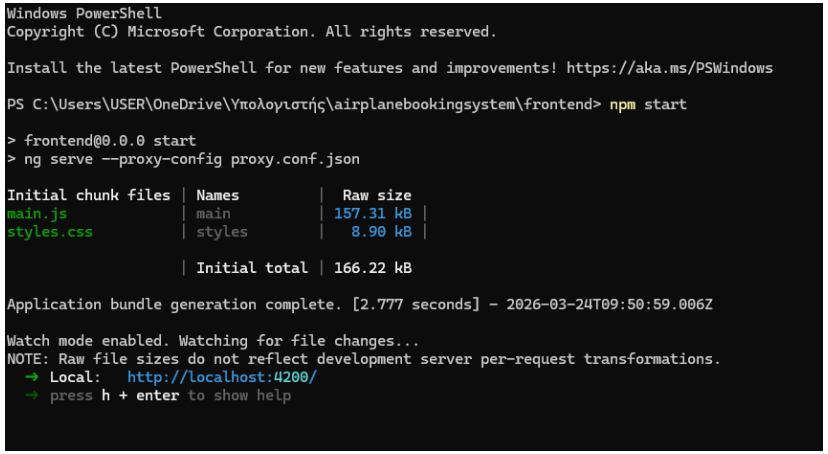


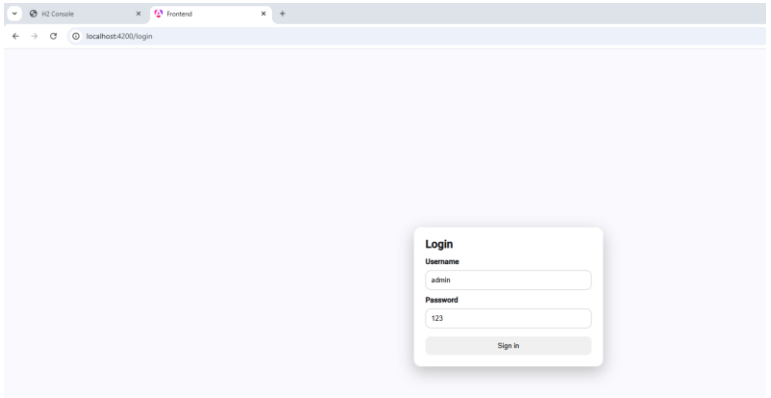
# APPLICATION RUN INSTRUCTIONS

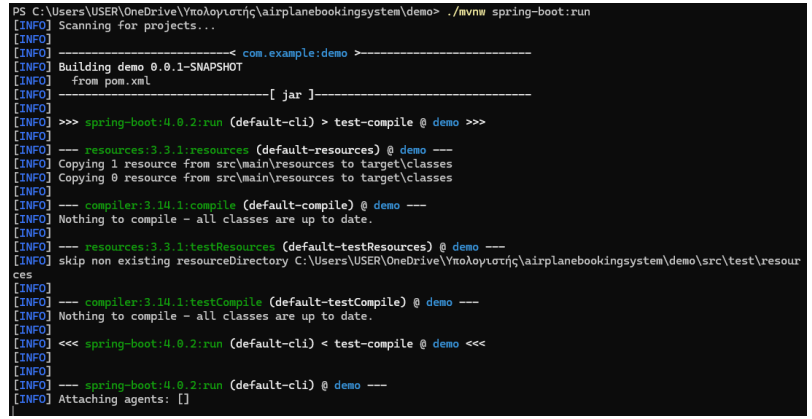


# RUN SCREENS


H2 Console
Frontend
localhost:8080/h2-console/login.jsp?jsessionid=2eab9c5ba84484492ea7214252b79289
English
Preferences
Tools
Help
Login
Saved Settings:
Generic H2 (Embedded)
Setting Name:
Generic H2 (Embedded)
Save
Remove
Driver Class:
org.h2.Driver
JDBC URL:
jdbc:h2:mem:testdb
User Name:
sa
Password:
Connect
Test Connection

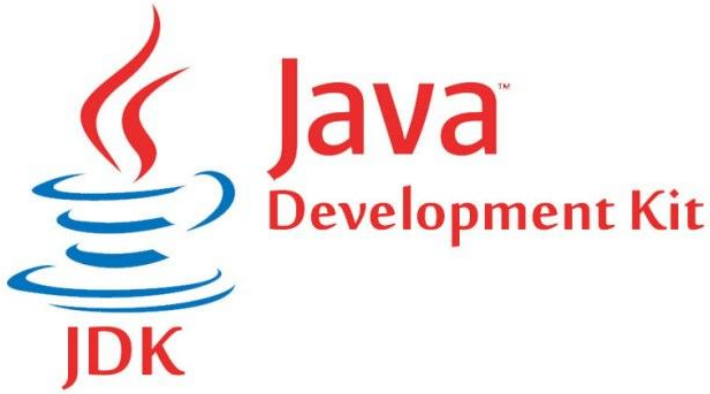

Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.
Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows
PS C:\Users\USER\OneDrive\Υπολογιστής\airplanebookingsystem\frontend> npm start
> frontend@0.0.0 start
> ng serve --proxy-config proxy.conf.json
Initial chunk files | Names | Raw size
main.js | main | 157.31 kB |
styles.css | styles | 8.90 kB |
| Initial total | 166.22 kB
Application bundle generation complete. [2.777 seconds] - 2026-03-24T09:50:59.006Z
Watch mode enabled. Watching for file changes...
NOTE: Raw file sizes do not reflect development server per-request transformations.
➜ Local: http://localhost:4200/
➜ press h + enter to show help


Login
Username
admin
Password
123
Sign in


PS C:\Users\USER\OneDrive\Υπολογιστής\airplanebookingsystem\demo> ./mvnw spring-boot:run
[INFO] Scanning for projects...
[INFO] ---------------------< com.example:demo >---------------------
[INFO] Building demo 0.0.1-SNAPSHOT
[INFO] from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO] >>> spring-boot:4.0.2:run (default-cli) > test-compile @ demo >>>
[INFO] --- resources:3.3.1:resources (default-resources) @ demo ---
[INFO] Copying 1 resource from src\main\resources to target\classes
[INFO] Copying 0 resource from src\main\resources to target\classes
[INFO] --- compiler:3.14.1:compile (default-compile) @ demo ---
[INFO] Nothing to compile - all classes are up to date.
[INFO] --- resources:3.3.1:testResources (default-testResources) @ demo ---
[INFO] skip non existing resourceDirectory C:\Users\USER\OneDrive\Υπολογιστής\airplanebookingsystem\demo\src\test\resources
[INFO] --- compiler:3.14.1:testCompile (default-testCompile) @ demo ---
[INFO] Nothing to compile - all classes are up to date.
[INFO] <<< spring-boot:4.0.2:run (default-cli) < test-compile @ demo <<<
[INFO] --- spring-boot:4.0.2:run (default-cli) @ demo ---
[INFO] Attaching agents: []









# Development Tools